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CLLAP: Contrastive Learning-based LiDAR-Augmented Pretraining for Enhanced Radar-Camera Fusion

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Abstract

*Accurate 3D object detection is critical for autonomous driving, necessitating reliable, cost-effective sensors capable of operating in adverse weather conditions. Camera and millimeter-wave radar fusion has emerged as a promising solution; however, these methods often rely on finely annotated radar data, which is scarce and labor-intensive to produce. To address this challenge, we present **CLLAP**, a **C**ontrastive **L**earning-based **LiDAR-A**ugmented **P**retraining framework that enhances the performance of existing radar-camera fusion methods for 3D object detection. CLLAP leverages abundant LiDAR data to generate pseudo-radar data using the proposed L2R (LiDAR-to-Radar) Sampling method. Then, it incorporates this data into a novel dual-stage, dual-modality contrastive learning strategy, enabling effective self-supervised learning from paired pseudo-radar and image data. This approach facilitates effective pretraining of existing radar-camera fusion models in a plug-and-play manner, enhancing their feature extraction capabilities and improving detection accuracy and robustness. Experimental results using NuScenes and Lyft Level 5 datasets demonstrate significant performance improvements across three baseline models, highlighting CLLAP's effectiveness in advancing radar-camera fusion for autonomous driving applications.*

1. Introduction

3D object detection plays a crucial role in autonomous driving by enhancing spatial understanding, situational awareness, and decision-making, enabling autonomous driving systems to navigate complex environments with precision and safety [21, 26]. However, relying on a single sensor can limit performance, prompting the adoption of multi-modality sensor fusion. By integrating data from cameras, radar, and LiDAR, multi-modality fusion offers a richer and more robust perception of the environment, significantly improving 3D detection accuracy. Among these approaches, camera-radar fusion is a cost-effective and resilient approach for real-world applications [36]. Unlike expensive and interference-prone LiDAR, radar offers reliable performance in adverse weather, making it ideal for intelligent transportation systems. This fusion delivers robust, affordable solutions for smarter, safer cities [8]. CRN [13] and RCBEVDet [19] are notable radar-camera fusion frameworks. CRN utilizes multi-modality deformable attention to generate a semantically rich bird's-eye-view representation from image features, while RCBEVDet leverages a dedicated cross-attention fusion framework to align radar and camera modalities in a unified space.

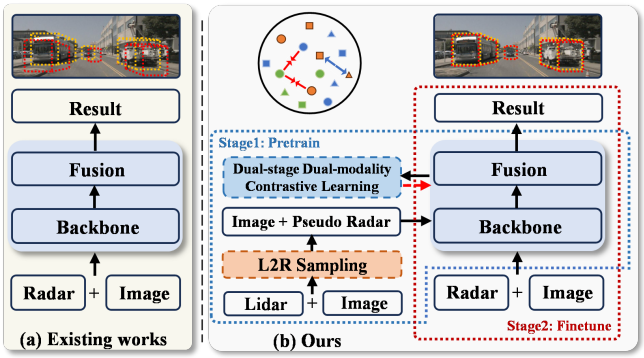


Figure 1. (a) Existing camera-radar fusion models rely solely on scarce, finely annotated radar-image data for training, leading to suboptimal performance. (b) The proposed CLLAP framework addresses this limitation by generating pseudo-radar data from large-scale unlabeled LiDAR data and employing a novel dual-stage, dual-modality contrastive learning strategy for self-supervised pretraining of the fusion model. Using the pretrained model as a starting point to train the fusion models in (a) significantly improves their detection performance and robustness.

Despite recent progress, further advancement in radar-based perception remains limited by the scarcity of large, well-annotated open-source radar datasets, which are crucial for both training and evaluation [4]. The intrinsic spar-

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sity and ambiguity of radar measurements also make manual annotation difficult and labor-intensive, further slowing dataset development [32]. Although several data augmentation strategies attempt to mitigate this limitation—such as rotating or duplicating radar point clouds, or transplanting objects across datasets to increase diversity—these operations often introduce non-negligible artifacts and inconsistencies. As a result, while such techniques may offer modest robustness gains, the errors they introduce ultimately limit their ability to address the fundamental challenge of radar data scarcity [27].

Recent LiDAR-guided knowledge distillation methods [2, 3, 35, 39] treat LiDAR as a privileged teacher and transfer its rich geometric cues to radar detectors through supervised distillation on paired, annotated radar–LiDAR data, thereby boosting radar-based 3D object detection. This raises a natural question: *can we exploit abundant LiDAR data in a label-free manner, by converting LiDAR into pseudo-radar measurements and using them to self-supervise radar–camera fusion pretraining, so that radar perception can be enhanced without relying on large-scale annotated radar–LiDAR datasets?*

To this end, we propose CLLAP, a novel **C**ontrastive **L**earning-based **L**iDAR-**A**ugmented **P**retraining framework. CLLAP generates pseudo-radar data from LiDAR data and incorporates a designed self-supervised contrastive learning strategy for effective model pretraining to enhance the sensing performance of existing radar-camera fusion models. Specifically, the CLLAP framework introduces two key innovations. First, L2R (LiDAR-to-Radar) Sampling generates high-quality pseudo-radar data from LiDAR, replicating radar-specific characteristics, including an estimated velocity distribution absent in the original LiDAR data. This method leverages large-scale, unlabeled LiDAR datasets to enrich training data diversity, enabling robust model training. Second, Dual-Stage Dual-Modality Contrastive Learning employs contrastive learning to align features from radar and camera modalities that represent the same object or scene. By combining cross-modal local and global contrastive losses with intra-modal global contrastive loss, this strategy enables robust self-supervised learning. Pretraining on unlabeled images and pseudo-radar data enhances feature alignment and representation across modalities, leveraging their complementary strengths. Together, these innovations reduce reliance on large paired radar-camera datasets, improve feature learning, and significantly enhance the performance and robustness of radar-camera fusion models. The main contributions are summarized as follows:

- We introduce **CLLAP**, a novel Contrastive Learning-based LiDAR-Augmented Pretraining framework that leverages large-scale pseudo-radar data generated from high-quality, open-source LiDAR datasets for self-supervised pretraining, addressing the scarcity of radar data and enhancing the performance of existing radar-camera fusion models.
- We develop the L2R Sampling method to generate high-quality pseudo-radar data from LiDAR and propose a Dual-Stage Dual-Modality Contrastive Learning strategy, improving the model's representation capabilities by carefully aligning and enhancing complementary features from radar and camera modalities.
- Extensive experiments on the NuScenes and Lyft Level 5 datasets demonstrate that our proposed pretraining framework consistently enhances the performance of three different baseline radar-camera fusion models.

2. Related Work

2.1. Camera-radar fusion 3d object detection

Accurate 3D object detection using radar is challenging due to the sparse radar point cloud and limited 3D semantic information. To overcome these limitations, fusion approaches combining radar and camera data have been explored. CRAFT [12] utilizes early fusion to integrate spatial and contextual features. ClusterFusion [14] preserves local spatial features through point cloud clustering, minimizing information loss. MVFusion [34] employs a Semantically Aligned Radar Encoder and a Radar-Guided Fusion Transformer for enhanced global correlation. BEV-radar [40] integrates data bi-directionally in the BEV domain, while BEVCar [28] uses attention-based enhancement initialized by sparse radar points to refine image representation. Unlike the above methods trained only on radar datasets, our approach leverages LiDAR data for pretraining, leading to improved performance.

2.2. Data Augmentation

To further enhance the performance of fusion-based detection, many studies have proposed data augmentation techniques to increase the diversity and robustness of training data. PasteFusion [37] introduces the Paste-Aug algorithm, which improves image and point cloud data while addressing occlusion issues and reducing computational demands. Deformable Feature Fusion Network [9] further advances 3D object detection by incorporating the Feature Alignment Transform (FAT) module, which mitigates feature misalignment in point cloud enhancement by recording transformation matrices and inverting voxel coordinates. C-CLOCs [38] improves multi-modality 3D object detection through Multimodal Ground Truth Sampling (MGS), which automates database generation via instance segmentation. Our L2R Sampling method enhances generalization by sampling real LiDAR data matching the radar point cloud distribution, rather than generating synthetic data for augmentation.

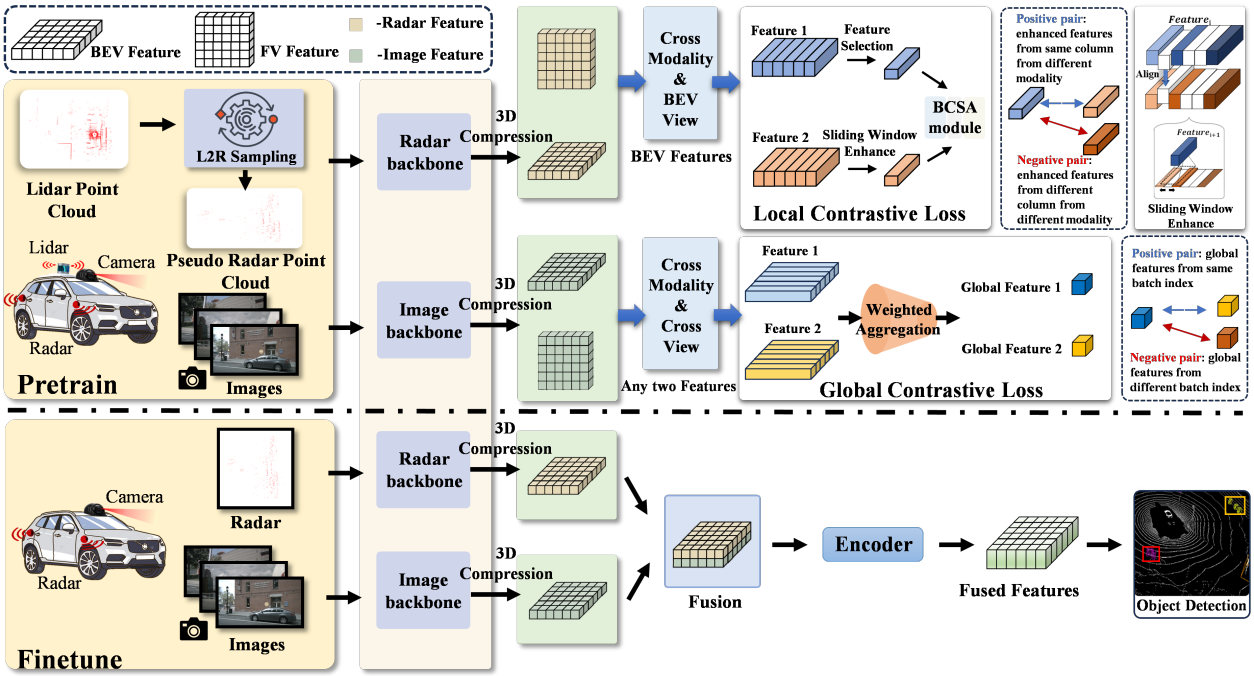


Figure 2. The overall architecture of the proposed CLLAP Method. The framework begins by generating pseudo-radar point clouds from LiDAR data using the L2R Sampling method. Combined with image data, the backbone processes these inputs to extract features, pre-trained using the proposed dual-stage, dual-modality contrastive learning strategy. The model is then fine-tuned with real radar data for 3D object detection task.

2.3. Contrastive Learning

Self-supervised learning (SSL) leverages pseudo-labels to generate representations for downstream tasks, with contrastive learning emerging as a key technique in computer vision [17]. In multi-modality 3D object detection, innovative SSL methods have advanced the field. CrossPoint [1] employs intra- and cross-modality losses to improve 3D point cloud understanding. ContrastAlign [29] aligns heterogeneous modalities for enhanced cross-modality 3D detection. CLIX^{3D} [10] achieves domain-invariant feature learning with supervised contrastive learning. MODCL [15] integrates point cloud-image mapping and supervised contrastive learning, surpassing traditional multi-modality methods on the SUN RGB-D dataset. In contrast to the aforementioned contrastive learning schemes, our method takes full advantage of the fusion models by applying multi-modality contrastive learning at both local and global stages.

2.4. Cross-Modal Knowledge Distillation

Cross-modal knowledge distillation (KD) leverages LiDAR as a privileged teacher to improve radar-based 3D perception. SCKD [35], RadarDistill [2], CRKD [39], and RCT-Distill [3] all distill rich LiDAR (or LiDAR–camera) representations into radar or radar–camera students via supervised feature and proposal alignment on paired, labeled radar–LiDAR data, consistently boosting detection performance. However, these methods inherently depend on large-scale annotated radar–LiDAR corpora and treat LiDAR only as a teacher whose predictions or intermediate features serve as distillation targets for specific radar or radar–camera samples. In contrast, our CLLAP framework does not perform supervised distillation from LiDAR outputs; instead, it treats large-scale *LiDAR-only* point clouds as an unlabeled geometric corpus, converts them into pseudo-radar measurements, and pairs them with images to conduct self-supervised contrastive pretraining of radar–camera fusion models, thereby improving radar perception without paired radar–LiDAR data or detection labels.

3. The Proposed Method

3.1. Overview

Fig. 2 illustrates the architecture of our proposed unsupervised contrastive learning method CLLAP, comprising two stages: pre-training and fine-tuning, divided into three steps. In the first step, the L2R Sampling module generates pseudo-radar data from large-scale LiDAR data, enabling initial pre-training of the feature encoder to capture diverse feature representations. The second step employs real radar data for secondary pre-training, enhancing the model's understanding of real-world radar data distribution. Together, these steps form the pre-training stage, leveraging a dual-stage, dual-modality contrastive learning approach to cap-

ture intrinsic correlations across modalities and views. In the fine-tuning stage, the third step uses real radar data for task-specific optimization. Section 3.2 details the L2R Sampling module, while Section 3.3 discusses the Dual-Stage Dual-Modality Contrastive Learning Strategy.

3.2. L2R Sampling module

In this section, we introduce the design of our L2R Sampling module, illustrated in Fig. 3. Unlike prior neural network-based methods for converting LiDAR to radar data [18, 30], our approach emulates radar point cloud distribution characteristics without training neural networks. This method bridges the distribution gap between LiDAR and radar data, enabling the effective use of abundant LiDAR datasets for radar-related tasks. The pseudo-radar data generation process comprises five steps: Gaussian Mixture Model (GMM) fitting, point cloud redundancy removal, two-stage sampling, velocity augmentation, and radar plane mapping.

The first step in sampling is GMM fitting. Radar point clouds in traffic scenarios exhibit highly non-uniform densities—sparse in most frames with occasional dense clusters. To realistically capture these patterns without relying on artificially uniform data, we model the density distribution using a GMM. Experiments (Appendix) show that variations in GMM parameters have minimal impact on our framework.

The second step is to use KD-tree-based spatial indexing method to remove redundancy. For each point, we calculate its minimum distance to neighboring points d_{min}, removing all points whose distance to other points is below a threshold $d_{\text{threshold}}$. The resulting point cloud, after redundancy removal, is sparser, preventing the concentration of points in any specific region, and better simulates the characteristics of radar point clouds.

In order to effectively simulate the sparse distribution characteristics of the radar point cloud in space, we perform the third step: two-stage sampling. LiDAR point clouds are typically denser in the center region, whereas radar point clouds exhibit a more uniform distribution across the entire space. To reflect this, we apply a weight-based sampling method to sample point clouds from both the non-central region and the global space. Based on the point cloud count N derived from the GMM fitting, we divide it equally into two sampling categories: the non-central region sample count N_1 and the global region sample count N_2, where $N = N_1 + N_2$. First, we sample N_1 point clouds located farther than a certain threshold (set to 15 meters in our experiments) from the center to obtain the external sampled point clouds. Then, the remaining points N_2 are sampled from the global region. This approach ensures that the sampled point clouds more accurately reflect the distribution characteristics of radar point clouds. During sampling, we incorporate point cloud intensity, sparsity, and distance to compute a weighted sampling weight for each point. The intensity weight w_{int} reflects the relative contribution of each point based on its point intensity. To ensure points with greater reflective intensity are properly emphasized during sampling, greater weights are assigned to them. The intensity weight for each point is calculated as $w_{\text{int}} = I_i^{\frac{1}{2}} / \sum_j I_j^{\frac{1}{2}}$, where I_i is the intensity of point i. To ensure accurate feature representation, we assign weights based on point sparsity and distance. Sparse regions require greater sampling to capture their structural and semantic information, quantified by the sparsity weight $w_{\text{spa}} = \sum_j D_{ij}^2$, where D_{ij} is the Euclidean distance between point i and its j-th nearest neighbor. Similarly, points farther from the center of the point cloud, often sparse in radar sensing, are assigned a distance weight $w_{\text{dist}} = 1/D_{iO}^2$, where D_{iO} is the Euclidean distance from point i to the origin. Finally, the overall sampling weight is a linear combination of the three individual weights, with scaling factors α_{int}, α_{spa}, α_{dist} controlling the contributions of each weight: $w_{\text{final}} = \alpha_{\text{int}} w_{\text{int}} + \alpha_{\text{dist}} w_{\text{dist}} + \alpha_{\text{spa}} w_{\text{spa}}$, where $\alpha_{int} = 4$, $\alpha_{spa} = 2$, $\alpha_{dist} = 4$ in our experiment (appendix). By normalizing w_{final} for all points, we ensure the sampling probabilities are well-defined.

To supplement the missing velocity information in the LiDAR data and align the motion between frames, in the fourth step, we use FlowNet3D [20] to predict the scene flow between consecutive frames, completing the velocity information augmentation step. Using the FlowNet3D model, we can predict velocity information for each LiDAR point, further enhancing the motion characteristics of the generated pseudo-radar point clouds.

Since radar usually does not provide accurate altitude information, we map the point cloud to the radar plane in the final step of radar plane mapping. Due to the random nature of the sampling itself, our method can generate pseudo-radar point clouds with different distributions for the same scene, thus providing diverse datasets for model training and further enhancing the generalization ability of the model. The visual results of the L2R operation are shown in the Fig. 4.

3.3. Dual-Stage Dual-Modality Contrastive Learning Strategy

Given the scarcity of open-source datasets and the challenges associated with labeling 3D radar point clouds, further improvements in model performance are limited by the available data. To address this, we propose leveraging unsupervised contrastive learning to pre-train the model using unlabeled radar data and pseudo-radar point clouds generated through the L2R Sampling module. This pre-training process aims to enhance model performance by using a contrastive learning framework that incorporates both Lo-

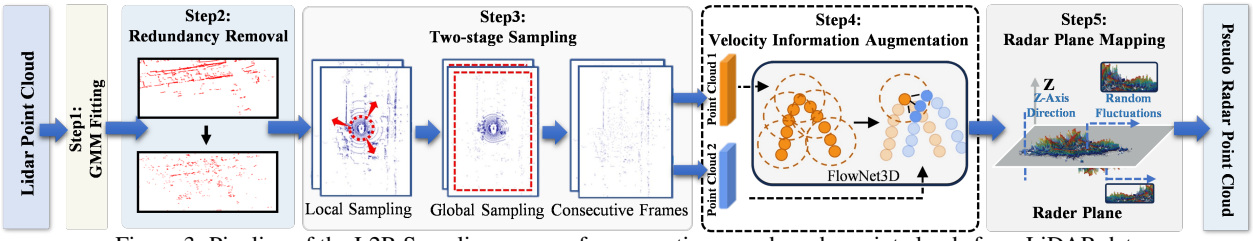


Figure 3. Pipeline of the L2R Sampling process for generating pseudo-radar point clouds from LiDAR data.

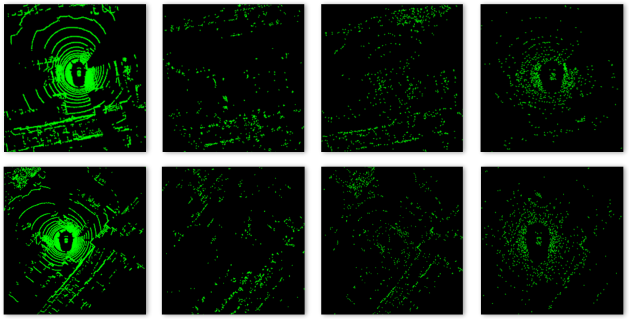


Figure 4. Comparison of sampling methods visualization. This figure illustrates different point cloud sampling methods: (a) represents the original LiDAR point cloud frames, (b) shows the ground-truth radar point cloud frames, (c) displays the pseudo-radar point cloud frames generated by the L2R Sampling module, and (d) depicts the point cloud frames sampled with distance as the weighting factor.

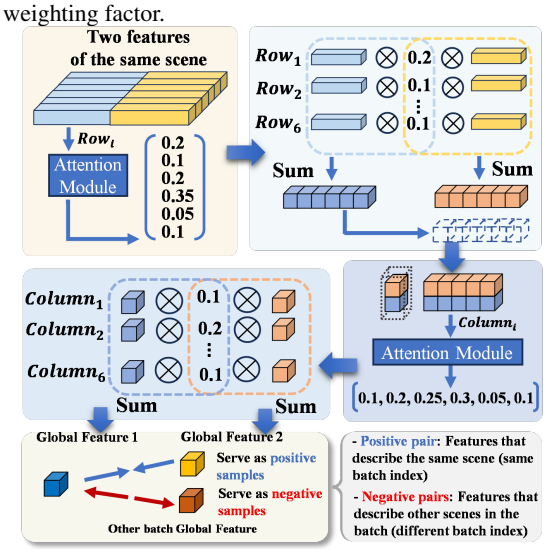


Figure 5. An overall framework for Global Contrastive Loss. The inputs consist of features from multiple views across different modalities, grouped into six sets of contrast losses. An attention mechanism is applied to perform weighted aggregation of features into global features, followed by the computation of contrast losses.

cal Contrastive Loss and Global Contrastive Loss.

3.3.1. Local Contrastive Loss.

Different modalities, such as radar and image, often provide complementary information. However, even when their feature maps are structurally aligned—such as both being represented in the bird's eye view (BEV) format—significant semantic gaps can emerge due to inherent differences in their sensory characteristics.

To address this issue, we align corresponding feature columns across modalities to substantially reduce the semantic gap and enhance the quality of fused features. By aligning column features, we can directly optimize localized feature correspondence, which is particularly crucial for tasks requiring spatial precision, such as object detection and localization. The reason for choosing a column feature for the loss function stems from the observation that corresponding columns across different modalities and views represent the same underlying features. This alignment is maintained across views, underscoring the importance of columns as units of correspondence for optimization.

Building on this insight, we propose the Local Contrastive Loss as shown in Fig. 2, which enforces alignment between feature columns across different modalities. To implement this, we first extract the BEV feature based on the baseline model from the radar (F_{rad}) and the camera (F_{img}) modalities. Select N (number of batch size) feature columns from each modality. Each set of corresponding columns represents the same spatial position across the modalities. For fine-grained feature misalignment, we conduct contrastive learning on carefully designed positive sample pairs. Specifically, for each feature column, we define a local search area with a width of $R(R = 5)$ in the query feature map and use a sliding window of width $r(r = 3, r<R)$ to slide through the local search area to extract $n = R - r + 1$ candidate areas. Each candidate area is adaptively aggregated via an attention module to match the feature column's dimensions, followed by pairwise similarity comparison. The highest-scoring candidate is selected as a positive pair, effectively replacing direct positional alignment with a learned, similarity-driven matching strategy. This approach enhances feature correspondence precision while maintaining robustness to spatial discrepancies between modalities. Furthermore, we introduce a Bidirectional Channel-Spatial Attention module, as shown in Fig. 6. For calculated positive pairs, their channel and spatial dimensions of the feature maps contain distinct semantic meanings. This inspired us to decoupling the modeling of channel and spatial dimensions, allowing the model to conduct finer-grained feature align-

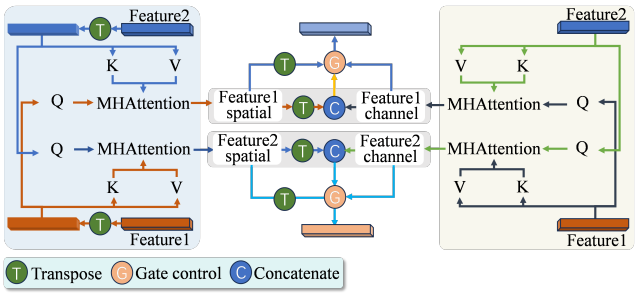


Figure 6. The overall architecture of the proposed BCSA Module.

ment. Specifically, our module independently applies attention mechanisms along the channel and spatial dimensions. *Channel Attention* learns importance weights for different feature channels, highlighting those most discriminative for the current task. *Spatial Attention* learns importance weights for different spatial locations within the feature column, emphasizing the importance of the region of interest in object detection while suppressing interference from background regions. Through this bidirectional decoupling and weighting across channel and spatial dimensions, the module adaptively refines the most relevant feature information for positive pair contrastive learning, suppressing irrelevant noise and guiding the model to learn more discriminative feature representations.

We use InfoNCE [24] loss to minimize the distance between positive pairs and maximize the distance from negative pairs:

$$\mathcal{L}_{\text{col}} = -\frac{1}{N}\sum_{i=1}^{N}\log\left(\frac{\exp\left(\text{sim}\left(\mathbf{f}_i^{\text{rad}}, \mathbf{f}_i^{\text{img}}\right)/\tau\right)}{\sum_{j=1}^{N}\exp\left(\text{sim}\left(\mathbf{f}_i^{\text{rad}}, \mathbf{f}_j^{\text{img}}\right)/\tau\right)}\right) \quad (1)$$

where $\text{sim}(\cdot,\cdot)$ denotes cosine similarity, and τ is a temperature parameter that controls the smoothness of the softmax distribution. This loss encourages the contents of the same columns in different modalities to be more similar and the contents of different columns to be less similar.

3.3.2. Global Contrastive Loss.

While Local Contrastive Loss effectively aligns corresponding features at a fine-grained level, it may lead to semantic disjunction between features across different columns. To address this, we introduce Global Contrastive Loss, as shown in Fig. 5, to better leverage multi-view and multi-modality representations.

In multi-modality feature fusion, the routine is to exploit complementary information from diverse sensory modalities (e.g., radar and camera). However, relying solely on simple modality-based fusion often overlooks the important variations that arise from multiple viewpoints. Each view offers unique insights that are crucial for forming a more complete and accurate representation of objects. By combining these views, we can enhance the model's ability to understand the scene from multiple dimensions, improving model robustness and accuracy. Therefore, we propose a novel approach that integrates both multi-modality and multi-view features, effectively combining their respective advantages to optimize feature extraction and fusion. This method not only bridges the semantic gaps between modalities but also ensures the model can capture the variations across different views, resulting in a more comprehensive and effective representation for downstream tasks.

Considering all views and modalities, our approach results in six sets of contrastive losses: $(F_{img}^{bev}, F_{img}^{fv})$, $(F_{img}^{bev}, F_{rad}^{fv})$, $(F_{img}^{bev}, F_{rad}^{bev})$, $(F_{img}^{fv}, F_{rad}^{bev})$, $(F_{img}^{fv}, F_{rad}^{fv})$, $(F_{rad}^{bev}, F_{rad}^{fv})$. Here, F^{bev} and F^{fv} represent bird's-eye view and front view(FV) features, respectively, while F_{img} and F_{rad} denote image and radar features, respectively. Before calculating the Global Contrastive Loss, features are further processed using an attention mechanism to compute weighted representations. For illustration, two feature representations, $(F_{img}^{bev}, F_{img}^{fv})$ are used as an example, with each having dimensions $C \times H \times W$, where H and W represent the spatial feature height and width, and C denotes the number of channels.

To integrate the features from both modalities, we apply a weighted summation approach. Instead of relying on attention weights derived from a single modality, which could lead to the loss of crucial information from the other modality, we first concatenate the features. This concatenation enables the attention module to simultaneously consider the feature distributions from both views, ensuring a more balanced and comprehensive weight distribution. For the first step of feature aggregation, shared attention weights are computed for the rows. These weights determine the importance of each row by comparing the contributions of both views in the concatenated feature representation. The row weights are then used to perform a weighted summation over the rows of the BEV and FV features, resulting in aggregated column-wise feature representations for each modality. This operation captures the key information across rows, producing features that are condensed along the row dimension. Next, similar to row aggregation, we calculate attention weights for the columns using the previously aggregated features. These column weights are computed based on the concatenated column-wise features of the two modalities. The column weights are then used to perform a weighted summation across columns, yielding the final global feature representations for both modalities. Through this two-step process of row-wise and column-wise aggregation, we achieve a global feature representation for each modality, F_{global}^{bev} and F_{global}^{fv}. These global features are now in a compact C-dimensional form that better characterizes the overall scene.

For a given feature map, positive samples correspond to the features from the same batch that represent the same scene. For example, for the contrastive loss $\mathcal{L}(F_{\text{img}}^{\text{bev}}, F_{\text{img}}^{\text{fv}})$, the positive samples are the feature from the same batch, denoted as $F_{\text{img}}^{\text{bev}(i)}$ and $F_{\text{img}}^{\text{fv}(i)}$, where i is the index of the scene

within the batch. Negative samples are those features that belong to different scenes within the same batch. The contrastive loss for $\mathcal{L}(F_{\text{img}}^{\text{bev}}, F_{\text{img}}^{\text{fv}})$ can be formulated as follows:

$$\mathcal{L}\left(F_{\text{img}}^{\text{bev}}, F_{\text{img}}^{\text{fv}}\right) = -\log \frac{\exp\left(\text{sim}\left(F_{\text{global}}^{\text{bev}(i)}, F_{\text{global}}^{\text{fv}(i)}\right)/\tau\right)}{\sum_{k=1}^{N} \exp\left(\text{sim}\left(F_{\text{global}}^{\text{bev}(i)}, F_{\text{global}}^{\text{fv}(k)}\right)/\tau\right)} \tag{2}$$

The overall Global Contrastive Loss is then computed as the sum of the individual contrastive losses for all the positive and negative sample pairs across all sets of features:

$$\begin{aligned}\mathcal{L}_{\text{global}} = &\mathcal{L}\left(F_{\text{img}}^{\text{bev}}, F_{\text{img}}^{\text{fv}}\right) + \mathcal{L}\left(F_{\text{img}}^{\text{bev}}, F_{\text{rad}}^{\text{fv}}\right) + \mathcal{L}\left(F_{\text{img}}^{\text{fv}}, F_{\text{rad}}^{\text{fv}}\right) \\ &+ \mathcal{L}\left(F_{\text{img}}^{\text{bev}}, F_{\text{rad}}^{\text{bev}}\right) + \mathcal{L}\left(F_{\text{img}}^{\text{fv}}, F_{\text{rad}}^{\text{bev}}\right) + \mathcal{L}\left(F_{\text{rad}}^{\text{bev}}, F_{\text{rad}}^{\text{fv}}\right)\end{aligned} \tag{3}$$

Finally, we experimentally verify that the weight of global contrastive loss is 1/6 of the best effect in the appendix, so our overall contrastive loss is: $\mathcal{L} = \mathcal{L}_{\text{global}}/6 + \mathcal{L}_{\text{col}}$.

4. Experiment

4.1. Implementation Details

Backbones: Our experiments used CRN, RCBEVDet, and BEVFusion [22] as backbones to evaluate our methods. CRN employs multi-modality deformable attention to transform image features into a semantically rich bird's-eye-view (BEV) representation. BEVFusion, originally designed for lidar-camera fusion, was adapted for radar-camera fusion to enable direct comparison. RCBEVDet integrates RadarBEVNet and a Cross-Attention Multi-layer Fusion module to align and merge radar and camera data. All backbones were configured using the best publicly available settings from their open-source implementations.
Dataset: Experiments utilized the NuScenes [5] and Lyft Level 5 [11] datasets, established benchmarks in autonomous driving. NuScenes provides 3D object annotations from a full sensor suite across diverse urban scenarios, while Lyft Level 5 includes over 1,000 hours of lidar and camera data tracking vehicles, cyclists, and pedestrians.
Metrics: To assess the performance of our methods, we used standard evaluation metrics, including mean Average Precision (mAP) and NuScenes Detection Score (NDS)[5]. These metrics provide a comprehensive assessment of detection accuracy and overall system performance.

4.2. Main Results

Tab. 1 summarizes the performance gains brought by our pretraining on the nuScenes validation set. On CRN, CLLAP yields an absolute improvement of +3.23 mAP and +2.28 NDS. BEVFusion* also benefits notably, with mAP and NDS increased by +3.52 and +2.94, respectively. Even the strong RCBEVDet baseline is further improved, achieving gains of +0.61 mAP and +0.24 NDS. Compared with state-of-the-art radar–camera fusion methods, models equipped with our pretraining framework achieve superior overall performance, indicating that CLLAP is both effective and backbone-agnostic. To further assess the generalization ability of our pretraining, we also evaluate these models under adverse-weather conditions; detailed per-condition results are reported in the appendix.

Table 1. Comparison of overall 3D object detection performance on the NuScenes validation set. BEVFusion* represents a modified model for radar-image fusion. Metrics include mean Average Precision (mAP) and NuScenes Detection Score (NDS) with pretraining improvements shown in parentheses. **(w. CLLAP)** indicates results using our proposed CLLAP framework.

Method	**mAP↑**	**NDS↑**
CenterFusion [23]	33.20	45.30
CRAFT [12]	41.10	51.70
RCBEV4d [41]	45.30	56.30
SOLOFusion [25]	42.70	53.40
StreamPETR [31]	43.20	54.00
BEVfusion* [22]	34.89	35.26
BEVfusion* **(w. CLLAP)**	38.41 (+3.52)	38.20 (+2.94)
CRN [13]	47.30	55.74
CRN **(w. CLLAP)**	**50.53 (+3.23)**	**58.02 (+2.28)**
RCBEVDet [19]	45.30	56.85
RCBEVDet **(w. CLLAP)**	45.91 (+0.61)	57.09 (+0.24)

4.3. L2R Sampling

To evaluate the effectiveness of the proposed L2R Sampling module, we conducted experiments using VoxelNet [42] as the benchmark. The baseline used the NuScenes radar training dataset, while the "sample by distance" setting employed pseudo-radar points sampled with distance-based weights. The L2R Sampling module generated pseudo-radar data using our proposed strategy, with the "map" setting further projecting sampled points onto the radar plane to emulate radar's altitude insensitivity.

As shown in Tab. 2, the L2R Sampling module consistently outperformed the baselines in mAP and NDS. (Results for different sampling methods are shown in the appendix.) These improvements stem from the module's ability to better preserve radar point cloud distributions and retain object-relevant points. Additionally, to quantify its similarity to real radar data, we calculated the Chamfer Distance [7]. Over 4000 frames, the L2R Sampling module achieved a mean Chamfer Distance of 112.16, compared to 120.1 for the "sample by distance" method, confirming its ability to better replicate radar point structures.

4.4. Ablation Study

We perform ablation studies on the NuScenes validation set using CRN as the backbone to evaluate the effectiveness of each setting of our proposed framework.
Impact of Contrastive Losses. Our method leverages two contrastive loss components during pretraining to enhance model performance. As shown in Tab. 3, each contrastive

Table 2. Comparison of L2R Sampling module 3D object detection performance on NuScenes val set.

Training Data	mAP (↑)	NDS (↑)
Real radar data	7.40	31.36
Sample by distance	11.25(+3.85)	32.32(+0.96)
L2R Sampling module	**12.73(+5.33)**	**33.49(+2.13)**
Sample by distance(map)	10.57(+3.17)	32.53(+1.17)
L2R Sampling module(map)	11.02(+3.62)	32.75(+1.39)

Table 3. Ablation of the components of Dual-Stage Dual-Modality Contrastive Learning Strategy.

Config	Local	Global	mAP(↑)	NDS(↑)
baseline	-	-	47.30	55.74
1	✓	-	49.72 (+2.42)	57.73 (+1.99)
2	-	✓	48.13 (+0.83)	56.28 (+0.54)
3	✓	✓	**50.53 (+3.23)**	**58.02 (+2.28)**

Table 4. Impact of Global Contrastive families in $\mathcal{L}_{\text{global}}$.

Config	Component	mAP ↑	NDS ↑
baseline	None	47.30	55.74
1	Same-mod. Cross-view	50.11(+2.81)	57.80(+2.06)
2	Cross-mod. Same-view	49.81(+2.51)	57.83(+2.09)
3	Cross-mod. Cross-view	50.18(+2.88)	58.11(+2.37)
4	All (three families)	**50.53(+3.23)**	**58.02(+2.28)**

Table 5. Comparison of 3D object detection performance across different datasets pretrain on NuScenes val datasets.

Config	Lyft	NuScenes	mAP(↑)	NDS(↑)
baseline	-	-	47.30	55.74
1	✓	-	48.84 (+1.54)	56.87 (+1.13)
2	-	✓	48.60 (+1.30)	56.37 (+0.63)
3	✓	✓	**50.53 (+3.23)**	**58.02 (+2.28)**

loss significantly improves model performance, with the combined losses yielding the highest gains. These results validate the effectiveness of both loss components in our contrastive learning framework.

Impact of Global Contrastive Losses. We decompose the global contrastive loss into three families and train three variants, each activating only one family under identical settings; Tab. 4 shows that all three consistently improve over the baseline. *Same-modality cross-view* improves view invariance and geometric consistency across viewpoints. *Cross-modality same-view* reduces the semantic gap between sensors under a shared projection, enabling smoother and better-calibrated fusion. *Cross-modality cross-view* enforces compositional alignment under joint changes of viewpoint and modality, promoting cycle-consistent features and mitigating parallax- and occlusion-induced discrepancies. Overall, each family provides a distinct yet complementary benefit, jointly validating $\mathcal{L}_{\text{global}}$ as a principled composition of global contrastive constraints.

Impact of Pretraining Datasets. To assess the impact of pretraining data volume, we conducted an ablation experiment, with results summarized in Tab. 5. Pretraining on either Lyft Level 5 or nuScenes improves the downstream performance on the nuScenes validation set, while Lyft-only pretraining yields slightly larger gains than nuScenes-only pretraining. Combining both datasets leads to the best performance.

Table 6. Ablation of the step components.

Config	Step 1	Step 2	Step 3	mAP(↑)	NDS(↑)
1	-	-	✓	47.30	55.74
2	✓	-	✓	49.53 (+2.23)	57.77 (+2.03)
3	-	✓	✓	49.11 (+1.81)	57.57 (+1.83)
4	✓	✓	✓	**50.53 (+3.23)**	**58.02 (+2.28)**

Impact of the BCSA Module. To isolate the benefit of BCSA beyond generic cross-attention, we replace it with a bi-directional cross-attention that applies a single multi-head attention along the spatial sequence, without channel–spatial decoupling or gated fusion. Under identical training, this vanilla variant yields 49.77 mAP / 57.57 NDS—only +0.08 mAP over removing the module entirely (49.69 / 57.74) while reducing NDS by 0.17—suggesting that naive cross-modal token mixing introduces noisy associations that do not improve overall driving quality. In contrast, BCSA attains 50.53 mAP / 58.02 NDS, indicating that its structured channel–spatial interactions and normalized, gated bi-directional fusion better exploit complementary cross-modal cues while suppressing spurious correlations, thereby improving both detection accuracy and the holistic NDS metric.

Impact of the Framework Steps. We evaluated the contribution of each pertaining step in our framework, as summarized in Tab. 6. Step 1, which involves pre-training on pseudo-radar point clouds, improves mAP and NDS. This indicates that the L2R Sampling module and contrastive learning enhance generalization. Step 2, using a small set of real radar data for secondary pre-training, further improves mAP and NDS, highlighting the value of real radar data for refining model accuracy. The best performance was achieved by combining all three steps, resulting in significant improvements across mAP and NDS. This demonstrates that integrating pseudo-radar data, real radar data, and fine-tuning yields the best performance.

5. Conclusion

In this paper, we propose CLLAP, a novel Contrastive Learning-based LiDAR-Augmented Pretraining framework designed to enhance the performance and robustness of existing radar-camera fusion methods. We introduce the L2R module, which generates high-quality pseudo-radar data from large-scale LiDAR datasets, enhancing data diversity and facilitating robust model training. Additionally, we present a Dual-Stage Dual-modality Contrastive Learning Strategy that leverages features from different modalities and views to enhance the model's representation capabilities. Extensive experiments on multiple datasets demonstrate the effectiveness of CLLAP and the superiority of its components.

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